MSG No. 230.147 $Ia'\bar{3}'d$ [Type III, cubic]

Table 1: Wyckoff site: `16a`, site symmetry: `.-3'.`

No.	position	mapping
1	$[0, 0, 0]$	`[1,5,6,37,41,42]`
2	$\left[0, 0, \frac{1}{2}\right]$	`[2,10,11,38,46,47]`
3	$\left[\frac{1}{2}, 0, 0\right]$	`[3,7,12,39,43,48]`
4	$\left[0, \frac{1}{2}, 0\right]$	`[4,8,9,40,44,45]`
5	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[13,18,23,25,30,35]`
6	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[14,15,19,26,27,31]`
7	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[16,17,21,28,29,33]`
8	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[20,22,24,32,34,36]`
9	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[49,53,54,85,89,90]`
10	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[50,58,59,86,94,95]`
11	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[51,55,60,87,91,96]`
12	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[52,56,57,88,92,93]`
13	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[61,66,71,73,78,83]`
14	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[62,63,67,74,75,79]`
15	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[64,65,69,76,77,81]`
16	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[68,70,72,80,82,84]`

Table 2: Wyckoff site: `16b`, site symmetry: `.32'`

No.	position	mapping
1	$\left[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}\right]$	`[1,5,6,32,34,36]`
2	$\left[\frac{1}{8}, \frac{7}{8}, \frac{3}{8}\right]$	`[2,10,11,28,29,33]`
3	$\left[\frac{3}{8}, \frac{1}{8}, \frac{7}{8}\right]$	`[3,7,12,25,30,35]`
4	$\left[\frac{7}{8}, \frac{3}{8}, \frac{1}{8}\right]$	`[4,8,9,26,27,31]`
5	$\left[\frac{1}{8}, \frac{3}{8}, \frac{5}{8}\right]$	`[13,18,23,87,91,96]`
6	$\left[\frac{5}{8}, \frac{1}{8}, \frac{3}{8}\right]$	`[14,15,19,88,92,93]`
7	$\left[\frac{3}{8}, \frac{5}{8}, \frac{1}{8}\right]$	`[16,17,21,86,94,95]`
8	$\left[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}\right]$	`[20,22,24,85,89,90]`
9	$\left[\frac{7}{8}, \frac{7}{8}, \frac{7}{8}\right]$	`[37,41,42,68,70,72]`
10	$\left[\frac{7}{8}, \frac{1}{8}, \frac{5}{8}\right]$	`[38,46,47,64,65,69]`
11	$\left[\frac{5}{8}, \frac{7}{8}, \frac{1}{8}\right]$	`[39,43,48,61,66,71]`
12	$\left[\frac{1}{8}, \frac{5}{8}, \frac{7}{8}\right]$	`[40,44,45,62,63,67]`
13	$\left[\frac{5}{8}, \frac{5}{8}, \frac{5}{8}\right]$	`[49,53,54,80,82,84]`
14	$\left[\frac{5}{8}, \frac{3}{8}, \frac{7}{8}\right]$	`[50,58,59,76,77,81]`
15	$\left[\frac{7}{8}, \frac{5}{8}, \frac{3}{8}\right]$	`[51,55,60,73,78,83]`
16	$\left[\frac{3}{8}, \frac{7}{8}, \frac{5}{8}\right]$	`[52,56,57,74,75,79]`

Table 3: Wyckoff site: `24c`, site symmetry: `2.2'2'`

No.	position	mapping
1	$\left[\frac{1}{8}, 0, \frac{1}{4}\right]$	`[1,2,33,34]`
2	$\left[\frac{3}{8}, 0, \frac{3}{4}\right]$	`[3,25,52,74]`

continued ...

Table 3

No.	position	mapping
3	$\left[\frac{7}{8}, \frac{1}{2}, \frac{1}{4}\right]$	[4,26,51,73]
4	$\left[\frac{1}{4}, \frac{1}{8}, 0\right]$	[5,12,30,32]
5	$\left[0, \frac{1}{4}, \frac{1}{8}\right]$	[6,9,27,36]
6	$\left[\frac{1}{2}, \frac{1}{4}, \frac{7}{8}\right]$	[7,35,59,76]
7	$\left[\frac{3}{4}, \frac{3}{8}, 0\right]$	[8,31,58,77]
8	$\left[\frac{1}{4}, \frac{7}{8}, \frac{1}{2}\right]$	[10,29,56,79]
9	$\left[0, \frac{3}{4}, \frac{3}{8}\right]$	[11,28,55,83]
10	$\left[\frac{1}{8}, \frac{1}{2}, \frac{3}{4}\right]$	[13,40,62,87]
11	$\left[\frac{5}{8}, 0, \frac{1}{4}\right]$	[14,39,61,88]
12	$\left[\frac{1}{2}, \frac{1}{4}, \frac{3}{8}\right]$	[15,24,90,93]
13	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{8}\right]$	[16,43,71,95]
14	$\left[\frac{1}{4}, \frac{5}{8}, 0\right]$	[17,44,67,94]
15	$\left[\frac{1}{4}, \frac{3}{8}, \frac{1}{2}\right]$	[18,20,89,96]
16	$\left[\frac{3}{4}, \frac{1}{8}, \frac{1}{2}\right]$	[19,46,65,92]
17	$\left[\frac{3}{8}, \frac{1}{2}, \frac{1}{4}\right]$	[21,22,85,86]
18	$\left[0, \frac{1}{4}, \frac{5}{8}\right]$	[23,47,64,91]
19	$\left[\frac{7}{8}, 0, \frac{3}{4}\right]$	[37,38,69,70]
20	$\left[\frac{3}{4}, \frac{7}{8}, 0\right]$	[41,48,66,68]
21	$\left[0, \frac{3}{4}, \frac{7}{8}\right]$	[42,45,63,72]
22	$\left[\frac{5}{8}, \frac{1}{2}, \frac{3}{4}\right]$	[49,50,81,82]
23	$\left[\frac{3}{4}, \frac{5}{8}, \frac{1}{2}\right]$	[53,60,78,80]
24	$\left[\frac{1}{2}, \frac{3}{4}, \frac{5}{8}\right]$	[54,57,75,84]

Table 4: Wyckoff site: 24d, site symmetry: -4..

No.	position	mapping
1	$\left[\frac{3}{8}, 0, \frac{1}{4}\right]$	[1,2,14,61]
2	$\left[\frac{1}{8}, 0, \frac{3}{4}\right]$	[3,52,69,70]
3	$\left[\frac{5}{8}, \frac{1}{2}, \frac{1}{4}\right]$	[4,21,22,51]
4	$\left[\frac{1}{4}, \frac{3}{8}, 0\right]$	[5,12,17,67]
5	$\left[0, \frac{1}{4}, \frac{3}{8}\right]$	[6,9,23,64]
6	$\left[\frac{1}{2}, \frac{1}{4}, \frac{5}{8}\right]$	[7,15,24,59]
7	$\left[\frac{3}{4}, \frac{1}{8}, 0\right]$	[8,58,66,68]
8	$\left[\frac{1}{4}, \frac{5}{8}, \frac{1}{2}\right]$	[10,18,20,56]
9	$\left[0, \frac{3}{4}, \frac{1}{8}\right]$	[11,55,63,72]
10	$\left[\frac{7}{8}, \frac{1}{2}, \frac{3}{4}\right]$	[13,49,50,62]
11	$\left[\frac{1}{2}, \frac{3}{4}, \frac{7}{8}\right]$	[16,54,57,71]
12	$\left[\frac{3}{4}, \frac{7}{8}, \frac{1}{2}\right]$	[19,53,60,65]
13	$\left[\frac{5}{8}, 0, \frac{3}{4}\right]$	[25,37,38,74]
14	$\left[\frac{1}{8}, \frac{1}{2}, \frac{1}{4}\right]$	[26,73,85,86]
15	$\left[0, \frac{1}{4}, \frac{7}{8}\right]$	[27,36,47,91]
16	$\left[0, \frac{3}{4}, \frac{5}{8}\right]$	[28,42,45,83]
17	$\left[\frac{1}{4}, \frac{1}{8}, \frac{1}{2}\right]$	[29,79,89,96]
18	$\left[\frac{1}{4}, \frac{7}{8}, 0\right]$	[30,32,44,94]
19	$\left[\frac{3}{4}, \frac{5}{8}, 0\right]$	[31,41,48,77]

continued ...

Table 4

No.	position	mapping
20	$\left[\frac{7}{8}, 0, \frac{1}{4}\right]$	[33,34,39,88]
21	$\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{8}\right]$	[35,76,90,93]
22	$\left[\frac{3}{8}, \frac{1}{2}, \frac{3}{4}\right]$	[40,81,82,87]
23	$\left[\frac{1}{2}, \frac{3}{4}, \frac{3}{8}\right]$	[43,75,84,95]
24	$\left[\frac{3}{4}, \frac{3}{8}, \frac{1}{2}\right]$	[46,78,80,92]

Table 5: Wyckoff site: 32e, site symmetry: .3.

No.	position	mapping
1	$[x, x, x]$	[1,5,6]
2	$\left[x, -x, \frac{1}{2} - x\right]$	[2,10,11]
3	$\left[\frac{1}{2} - x, x, -x\right]$	[3,7,12]
4	$\left[-x, \frac{1}{2} - x, x\right]$	[4,8,9]
5	$\left[\frac{1}{4} - x, x + \frac{1}{4}, \frac{3}{4} - x\right]$	[13,18,23]
6	$\left[\frac{3}{4} - x, \frac{1}{4} - x, x + \frac{1}{4}\right]$	[14,15,19]
7	$\left[x + \frac{1}{4}, \frac{3}{4} - x, \frac{1}{4} - x\right]$	[16,17,21]
8	$\left[x + \frac{1}{4}, x + \frac{1}{4}, x + \frac{1}{4}\right]$	[20,22,24]
9	$\left[x + \frac{1}{4}, \frac{1}{4} - x, x + \frac{3}{4}\right]$	[25,30,35]
10	$\left[x + \frac{3}{4}, x + \frac{1}{4}, \frac{1}{4} - x\right]$	[26,27,31]
11	$\left[\frac{1}{4} - x, x + \frac{3}{4}, x + \frac{1}{4}\right]$	[28,29,33]
12	$\left[\frac{1}{4} - x, \frac{1}{4} - x, \frac{1}{4} - x\right]$	[32,34,36]
13	$[-x, -x, -x]$	[37,41,42]
14	$\left[-x, x, x + \frac{1}{2}\right]$	[38,46,47]
15	$\left[x + \frac{1}{2}, -x, x\right]$	[39,43,48]
16	$\left[x, x + \frac{1}{2}, -x\right]$	[40,44,45]
17	$\left[x + \frac{1}{2}, x + \frac{1}{2}, x + \frac{1}{2}\right]$	[49,53,54]
18	$\left[x + \frac{1}{2}, \frac{1}{2} - x, -x\right]$	[50,58,59]
19	$\left[-x, x + \frac{1}{2}, \frac{1}{2} - x\right]$	[51,55,60]
20	$\left[\frac{1}{2} - x, -x, x + \frac{1}{2}\right]$	[52,56,57]
21	$\left[\frac{3}{4} - x, x + \frac{3}{4}, \frac{1}{4} - x\right]$	[61,66,71]
22	$\left[\frac{1}{4} - x, \frac{3}{4} - x, x + \frac{3}{4}\right]$	[62,63,67]
23	$\left[x + \frac{3}{4}, \frac{1}{4} - x, \frac{3}{4} - x\right]$	[64,65,69]
24	$\left[x + \frac{3}{4}, x + \frac{3}{4}, x + \frac{3}{4}\right]$	[68,70,72]
25	$\left[x + \frac{3}{4}, \frac{3}{4} - x, x + \frac{1}{4}\right]$	[73,78,83]
26	$\left[x + \frac{1}{4}, x + \frac{3}{4}, \frac{3}{4} - x\right]$	[74,75,79]
27	$\left[\frac{3}{4} - x, x + \frac{1}{4}, x + \frac{3}{4}\right]$	[76,77,81]
28	$\left[\frac{3}{4} - x, \frac{3}{4} - x, \frac{3}{4} - x\right]$	[80,82,84]
29	$\left[\frac{1}{2} - x, \frac{1}{2} - x, \frac{1}{2} - x\right]$	[85,89,90]
30	$\left[\frac{1}{2} - x, x + \frac{1}{2}, x\right]$	[86,94,95]
31	$\left[x, \frac{1}{2} - x, x + \frac{1}{2}\right]$	[87,91,96]
32	$\left[x + \frac{1}{2}, x, \frac{1}{2} - x\right]$	[88,92,93]

Table 6: Wyckoff site: `48f`, site symmetry: `2..`

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{4}\right]$	`[1,2]`
2	$\left[\frac{1}{2}-x,\ 0,\ \frac{3}{4}\right]$	`[3,52]`
3	$\left[-x,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[4,51]`
4	$\left[\frac{1}{4},\ x,\ 0\right]$	`[5,12]`
5	$\left[0,\ \frac{1}{4},\ x\right]$	`[6,9]`
6	$\left[\frac{1}{2},\ \frac{1}{4},\ -x\right]$	`[7,59]`
7	$\left[\frac{3}{4},\ \frac{1}{2}-x,\ 0\right]$	`[8,58]`
8	$\left[\frac{1}{4},\ -x,\ \frac{1}{2}\right]$	`[10,56]`
9	$\left[0,\ \frac{3}{4},\ \frac{1}{2}-x\right]$	`[11,55]`
10	$\left[\frac{1}{4}-x,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[13,62]`
11	$\left[\frac{3}{4}-x,\ 0,\ \frac{1}{4}\right]$	`[14,61]`
12	$\left[\frac{1}{2},\ \frac{1}{4},\ x+\frac{1}{4}\right]$	`[15,24]`
13	$\left[\frac{1}{2},\ \frac{3}{4},\ \frac{1}{4}-x\right]$	`[16,71]`
14	$\left[\frac{1}{4},\ \frac{3}{4}-x,\ 0\right]$	`[17,67]`
15	$\left[\frac{1}{4},\ x+\frac{1}{4},\ \frac{1}{2}\right]$	`[18,20]`
16	$\left[\frac{3}{4},\ \frac{1}{4}-x,\ \frac{1}{2}\right]$	`[19,65]`
17	$\left[x+\frac{1}{4},\ \frac{1}{2},\ \frac{1}{4}\right]$	`[21,22]`
18	$\left[0,\ \frac{1}{4},\ \frac{3}{4}-x\right]$	`[23,64]`
19	$\left[x+\frac{1}{4},\ 0,\ \frac{3}{4}\right]$	`[25,74]`
20	$\left[x+\frac{3}{4},\ \frac{1}{2},\ \frac{1}{4}\right]$	`[26,73]`
21	$\left[0,\ \frac{1}{4},\ \frac{1}{4}-x\right]$	`[27,36]`
22	$\left[0,\ \frac{3}{4},\ x+\frac{1}{4}\right]$	`[28,83]`
23	$\left[\frac{1}{4},\ x+\frac{3}{4},\ \frac{1}{2}\right]$	`[29,79]`
24	$\left[\frac{1}{4},\ \frac{1}{4}-x,\ 0\right]$	`[30,32]`
25	$\left[\frac{3}{4},\ x+\frac{1}{4},\ 0\right]$	`[31,77]`
26	$\left[\frac{1}{4}-x,\ 0,\ \frac{1}{4}\right]$	`[33,34]`
27	$\left[\frac{1}{2},\ \frac{1}{4},\ x+\frac{3}{4}\right]$	`[35,76]`
28	$\left[-x,\ 0,\ \frac{3}{4}\right]$	`[37,38]`
29	$\left[x+\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	`[39,88]`
30	$\left[x,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[40,87]`
31	$\left[\frac{3}{4},\ -x,\ 0\right]$	`[41,48]`
32	$\left[0,\ \frac{3}{4},\ -x\right]$	`[42,45]`
33	$\left[\frac{1}{2},\ \frac{3}{4},\ x\right]$	`[43,95]`
34	$\left[\frac{1}{4},\ x+\frac{1}{2},\ 0\right]$	`[44,94]`
35	$\left[\frac{3}{4},\ x,\ \frac{1}{2}\right]$	`[46,92]`
36	$\left[0,\ \frac{1}{4},\ x+\frac{1}{2}\right]$	`[47,91]`
37	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	`[49,50]`
38	$\left[\frac{3}{4},\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[53,60]`
39	$\left[\frac{1}{2},\ \frac{3}{4},\ x+\frac{1}{2}\right]$	`[54,57]`
40	$\left[0,\ \frac{3}{4},\ x+\frac{3}{4}\right]$	`[63,72]`
41	$\left[\frac{3}{4},\ x+\frac{3}{4},\ 0\right]$	`[66,68]`
42	$\left[x+\frac{3}{4},\ 0,\ \frac{3}{4}\right]$	`[69,70]`
43	$\left[\frac{1}{2},\ \frac{3}{4},\ \frac{3}{4}-x\right]$	`[75,84]`
44	$\left[\frac{3}{4},\ \frac{3}{4}-x,\ \frac{1}{2}\right]$	`[78,80]`
45	$\left[\frac{3}{4}-x,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[81,82]`
46	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[85,86]`

continued ...

Table 6

No.	position	mapping
47	$\left[\frac{1}{4},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	[89,96]
48	$\left[\frac{1}{2},\ \frac{1}{4},\ \frac{1}{2}-x\right]$	[90,93]

Table 7: Wyckoff site: 48g, site symmetry: ..2'

No.	position	mapping
1	$\left[\frac{1}{8},\ y,\ \frac{1}{4}-y\right]$	[1,34]
2	$\left[\frac{1}{8},\ -y,\ y+\frac{1}{4}\right]$	[2,33]
3	$\left[\frac{3}{8},\ y,\ y+\frac{3}{4}\right]$	[3,25]
4	$\left[\frac{7}{8},\ \frac{1}{2}-y,\ \frac{1}{4}-y\right]$	[4,26]
5	$\left[\frac{1}{4}-y,\ \frac{1}{8},\ y\right]$	[5,32]
6	$\left[y,\ \frac{1}{4}-y,\ \frac{1}{8}\right]$	[6,36]
7	$\left[\frac{1}{2}-y,\ \frac{1}{4}-y,\ \frac{7}{8}\right]$	[7,35]
8	$\left[y+\frac{3}{4},\ \frac{3}{8},\ y\right]$	[8,31]
9	$\left[-y,\ y+\frac{1}{4},\ \frac{1}{8}\right]$	[9,27]
10	$\left[\frac{1}{4}-y,\ \frac{7}{8},\ \frac{1}{2}-y\right]$	[10,29]
11	$\left[y,\ y+\frac{3}{4},\ \frac{3}{8}\right]$	[11,28]
12	$\left[y+\frac{1}{4},\ \frac{1}{8},\ -y\right]$	[12,30]
13	$\left[\frac{1}{8},\ \frac{1}{2}-y,\ \frac{3}{4}-y\right]$	[13,87]
14	$\left[\frac{5}{8},\ y,\ y+\frac{1}{4}\right]$	[14,88]
15	$\left[y+\frac{1}{2},\ \frac{1}{4}-y,\ \frac{3}{8}\right]$	[15,93]
16	$\left[\frac{1}{2}-y,\ \frac{3}{4}-y,\ \frac{1}{8}\right]$	[16,95]
17	$\left[y+\frac{1}{4},\ \frac{5}{8},\ y\right]$	[17,94]
18	$\left[\frac{1}{4}-y,\ \frac{3}{8},\ y+\frac{1}{2}\right]$	[18,96]
19	$\left[\frac{3}{4}-y,\ \frac{1}{8},\ \frac{1}{2}-y\right]$	[19,92]
20	$\left[y+\frac{1}{4},\ \frac{3}{8},\ \frac{1}{2}-y\right]$	[20,89]
21	$\left[\frac{3}{8},\ y+\frac{1}{2},\ \frac{1}{4}-y\right]$	[21,86]
22	$\left[\frac{3}{8},\ \frac{1}{2}-y,\ y+\frac{1}{4}\right]$	[22,85]
23	$\left[y,\ y+\frac{1}{4},\ \frac{5}{8}\right]$	[23,91]
24	$\left[\frac{1}{2}-y,\ y+\frac{1}{4},\ \frac{3}{8}\right]$	[24,90]
25	$\left[\frac{7}{8},\ -y,\ y+\frac{3}{4}\right]$	[37,70]
26	$\left[\frac{7}{8},\ y,\ \frac{3}{4}-y\right]$	[38,69]
27	$\left[\frac{5}{8},\ -y,\ \frac{1}{4}-y\right]$	[39,61]
28	$\left[\frac{1}{8},\ y+\frac{1}{2},\ y+\frac{3}{4}\right]$	[40,62]
29	$\left[y+\frac{3}{4},\ \frac{7}{8},\ -y\right]$	[41,68]
30	$\left[-y,\ y+\frac{3}{4},\ \frac{7}{8}\right]$	[42,72]
31	$\left[y+\frac{1}{2},\ y+\frac{3}{4},\ \frac{1}{8}\right]$	[43,71]
32	$\left[\frac{1}{4}-y,\ \frac{5}{8},\ -y\right]$	[44,67]
33	$\left[y,\ \frac{3}{4}-y,\ \frac{7}{8}\right]$	[45,63]
34	$\left[y+\frac{3}{4},\ \frac{1}{8},\ y+\frac{1}{2}\right]$	[46,65]
35	$\left[-y,\ \frac{1}{4}-y,\ \frac{5}{8}\right]$	[47,64]
36	$\left[\frac{3}{4}-y,\ \frac{7}{8},\ y\right]$	[48,66]
37	$\left[\frac{5}{8},\ y+\frac{1}{2},\ \frac{3}{4}-y\right]$	[49,82]
38	$\left[\frac{5}{8},\ \frac{1}{2}-y,\ y+\frac{3}{4}\right]$	[50,81]
39	$\left[\frac{7}{8},\ y+\frac{1}{2},\ y+\frac{1}{4}\right]$	[51,73]

continued ...

Table 7

No.	position	mapping
40	$\left[\frac{3}{8}, -y, \frac{3}{4} - y\right]$	[52,74]
41	$\left[\frac{3}{4} - y, \frac{5}{8}, y + \frac{1}{2}\right]$	[53,80]
42	$\left[y + \frac{1}{2}, \frac{3}{4} - y, \frac{5}{8}\right]$	[54,84]
43	$\left[-y, \frac{3}{4} - y, \frac{3}{8}\right]$	[55,83]
44	$\left[y + \frac{1}{4}, \frac{7}{8}, y + \frac{1}{2}\right]$	[56,79]
45	$\left[\frac{1}{2} - y, y + \frac{3}{4}, \frac{5}{8}\right]$	[57,75]
46	$\left[\frac{3}{4} - y, \frac{3}{8}, -y\right]$	[58,77]
47	$\left[y + \frac{1}{2}, y + \frac{1}{4}, \frac{7}{8}\right]$	[59,76]
48	$\left[y + \frac{3}{4}, \frac{5}{8}, \frac{1}{2} - y\right]$	[60,78]

Table 8: Wyckoff site: 96h, site symmetry: 1

No.	position	mapping
1	$[x, y, z]$	[1]
2	$\left[x, -y, \frac{1}{2} - z\right]$	[2]
3	$\left[\frac{1}{2} - x, y, -z\right]$	[3]
4	$\left[-x, \frac{1}{2} - y, z\right]$	[4]
5	$[z, x, y]$	[5]
6	$[y, z, x]$	[6]
7	$\left[\frac{1}{2} - y, z, -x\right]$	[7]
8	$\left[-z, \frac{1}{2} - x, y\right]$	[8]
9	$\left[-y, \frac{1}{2} - z, x\right]$	[9]
10	$\left[z, -x, \frac{1}{2} - y\right]$	[10]
11	$\left[y, -z, \frac{1}{2} - x\right]$	[11]
12	$\left[\frac{1}{2} - z, x, -y\right]$	[12]
13	$\left[\frac{1}{4} - x, z + \frac{1}{4}, \frac{3}{4} - y\right]$	[13]
14	$\left[\frac{3}{4} - x, \frac{1}{4} - z, y + \frac{1}{4}\right]$	[14]
15	$\left[\frac{3}{4} - z, \frac{1}{4} - y, x + \frac{1}{4}\right]$	[15]
16	$\left[z + \frac{1}{4}, \frac{3}{4} - y, \frac{1}{4} - x\right]$	[16]
17	$\left[y + \frac{1}{4}, \frac{3}{4} - x, \frac{1}{4} - z\right]$	[17]
18	$\left[\frac{1}{4} - y, x + \frac{1}{4}, \frac{3}{4} - z\right]$	[18]
19	$\left[\frac{3}{4} - y, \frac{1}{4} - x, z + \frac{1}{4}\right]$	[19]
20	$\left[y + \frac{1}{4}, x + \frac{1}{4}, z + \frac{1}{4}\right]$	[20]
21	$\left[x + \frac{1}{4}, \frac{3}{4} - z, \frac{1}{4} - y\right]$	[21]
22	$\left[x + \frac{1}{4}, z + \frac{1}{4}, y + \frac{1}{4}\right]$	[22]
23	$\left[\frac{1}{4} - z, y + \frac{1}{4}, \frac{3}{4} - x\right]$	[23]
24	$\left[z + \frac{1}{4}, y + \frac{1}{4}, x + \frac{1}{4}\right]$	[24]
25	$\left[x + \frac{1}{4}, \frac{1}{4} - z, y + \frac{3}{4}\right]$	[25]
26	$\left[x + \frac{3}{4}, z + \frac{1}{4}, \frac{1}{4} - y\right]$	[26]
27	$\left[z + \frac{3}{4}, y + \frac{1}{4}, \frac{1}{4} - x\right]$	[27]
28	$\left[\frac{1}{4} - z, y + \frac{3}{4}, x + \frac{1}{4}\right]$	[28]
29	$\left[\frac{1}{4} - y, x + \frac{3}{4}, z + \frac{1}{4}\right]$	[29]
30	$\left[y + \frac{1}{4}, \frac{1}{4} - x, z + \frac{3}{4}\right]$	[30]
31	$\left[y + \frac{3}{4}, x + \frac{1}{4}, \frac{1}{4} - z\right]$	[31]
32	$\left[\frac{1}{4} - y, \frac{1}{4} - x, \frac{1}{4} - z\right]$	[32]

continued ...

Table 8

No.	position	mapping
33	$\left[\frac{1}{4}-x,\ z+\frac{3}{4},\ y+\frac{1}{4}\right]$	[33]
34	$\left[\frac{1}{4}-x,\ \frac{1}{4}-z,\ \frac{1}{4}-y\right]$	[34]
35	$\left[z+\frac{1}{4},\ \frac{1}{4}-y,\ x+\frac{3}{4}\right]$	[35]
36	$\left[\frac{1}{4}-z,\ \frac{1}{4}-y,\ \frac{1}{4}-x\right]$	[36]
37	$[-x,\ -y,\ -z]$	[37]
38	$\left[-x,\ y,\ z+\frac{1}{2}\right]$	[38]
39	$\left[x+\frac{1}{2},\ -y,\ z\right]$	[39]
40	$\left[x,\ y+\frac{1}{2},\ -z\right]$	[40]
41	$[-z,\ -x,\ -y]$	[41]
42	$[-y,\ -z,\ -x]$	[42]
43	$\left[y+\frac{1}{2},\ -z,\ x\right]$	[43]
44	$\left[z,\ x+\frac{1}{2},\ -y\right]$	[44]
45	$\left[y,\ z+\frac{1}{2},\ -x\right]$	[45]
46	$\left[-z,\ x,\ y+\frac{1}{2}\right]$	[46]
47	$\left[-y,\ z,\ x+\frac{1}{2}\right]$	[47]
48	$\left[z+\frac{1}{2},\ -x,\ y\right]$	[48]
49	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[49]
50	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ -z\right]$	[50]
51	$\left[-x,\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	[51]
52	$\left[\frac{1}{2}-x,\ -y,\ z+\frac{1}{2}\right]$	[52]
53	$\left[z+\frac{1}{2},\ x+\frac{1}{2},\ y+\frac{1}{2}\right]$	[53]
54	$\left[y+\frac{1}{2},\ z+\frac{1}{2},\ x+\frac{1}{2}\right]$	[54]
55	$\left[-y,\ z+\frac{1}{2},\ \frac{1}{2}-x\right]$	[55]
56	$\left[\frac{1}{2}-z,\ -x,\ y+\frac{1}{2}\right]$	[56]
57	$\left[\frac{1}{2}-y,\ -z,\ x+\frac{1}{2}\right]$	[57]
58	$\left[z+\frac{1}{2},\ \frac{1}{2}-x,\ -y\right]$	[58]
59	$\left[y+\frac{1}{2},\ \frac{1}{2}-z,\ -x\right]$	[59]
60	$\left[-z,\ x+\frac{1}{2},\ \frac{1}{2}-y\right]$	[60]
61	$\left[\frac{3}{4}-x,\ z+\frac{3}{4},\ \frac{1}{4}-y\right]$	[61]
62	$\left[\frac{1}{4}-x,\ \frac{3}{4}-z,\ y+\frac{3}{4}\right]$	[62]
63	$\left[\frac{1}{4}-z,\ \frac{3}{4}-y,\ x+\frac{3}{4}\right]$	[63]
64	$\left[z+\frac{3}{4},\ \frac{1}{4}-y,\ \frac{3}{4}-x\right]$	[64]
65	$\left[y+\frac{3}{4},\ \frac{1}{4}-x,\ \frac{3}{4}-z\right]$	[65]
66	$\left[\frac{3}{4}-y,\ x+\frac{3}{4},\ \frac{1}{4}-z\right]$	[66]
67	$\left[\frac{1}{4}-y,\ \frac{3}{4}-x,\ z+\frac{3}{4}\right]$	[67]
68	$\left[y+\frac{3}{4},\ x+\frac{3}{4},\ z+\frac{3}{4}\right]$	[68]
69	$\left[x+\frac{3}{4},\ \frac{1}{4}-z,\ \frac{3}{4}-y\right]$	[69]
70	$\left[x+\frac{3}{4},\ z+\frac{3}{4},\ y+\frac{3}{4}\right]$	[70]
71	$\left[\frac{3}{4}-z,\ y+\frac{3}{4},\ \frac{1}{4}-x\right]$	[71]
72	$\left[z+\frac{3}{4},\ y+\frac{3}{4},\ x+\frac{3}{4}\right]$	[72]
73	$\left[x+\frac{3}{4},\ \frac{3}{4}-z,\ y+\frac{1}{4}\right]$	[73]
74	$\left[x+\frac{1}{4},\ z+\frac{3}{4},\ \frac{3}{4}-y\right]$	[74]
75	$\left[z+\frac{1}{4},\ y+\frac{3}{4},\ \frac{3}{4}-x\right]$	[75]
76	$\left[\frac{3}{4}-z,\ y+\frac{1}{4},\ x+\frac{3}{4}\right]$	[76]
77	$\left[\frac{3}{4}-y,\ x+\frac{1}{4},\ z+\frac{3}{4}\right]$	[77]
78	$\left[y+\frac{3}{4},\ \frac{3}{4}-x,\ z+\frac{1}{4}\right]$	[78]
79	$\left[y+\frac{1}{4},\ x+\frac{3}{4},\ \frac{3}{4}-z\right]$	[79]

continued ...

Table 8

No.	position	mapping
80	$\left[\frac{3}{4}-y,\ \frac{3}{4}-x,\ \frac{3}{4}-z\right]$	[80]
81	$\left[\frac{3}{4}-x,\ z+\frac{1}{4},\ y+\frac{3}{4}\right]$	[81]
82	$\left[\frac{3}{4}-x,\ \frac{3}{4}-z,\ \frac{3}{4}-y\right]$	[82]
83	$\left[z+\frac{3}{4},\ \frac{3}{4}-y,\ x+\frac{1}{4}\right]$	[83]
84	$\left[\frac{3}{4}-z,\ \frac{3}{4}-y,\ \frac{3}{4}-x\right]$	[84]
85	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	[85]
86	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ z\right]$	[86]
87	$\left[x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[87]
88	$\left[x+\frac{1}{2},\ y,\ \frac{1}{2}-z\right]$	[88]
89	$\left[\frac{1}{2}-z,\ \frac{1}{2}-x,\ \frac{1}{2}-y\right]$	[89]
90	$\left[\frac{1}{2}-y,\ \frac{1}{2}-z,\ \frac{1}{2}-x\right]$	[90]
91	$\left[y,\ \frac{1}{2}-z,\ x+\frac{1}{2}\right]$	[91]
92	$\left[z+\frac{1}{2},\ x,\ \frac{1}{2}-y\right]$	[92]
93	$\left[y+\frac{1}{2},\ z,\ \frac{1}{2}-x\right]$	[93]
94	$\left[\frac{1}{2}-z,\ x+\frac{1}{2},\ y\right]$	[94]
95	$\left[\frac{1}{2}-y,\ z+\frac{1}{2},\ x\right]$	[95]
96	$\left[z,\ \frac{1}{2}-x,\ y+\frac{1}{2}\right]$	[96]